4. Merge Sort

Merge Sort is a divide-and-conquer algorithm that divides the list into halves, recursively sorts them, and then merges the sorted halves.

```
def merge_sort(data):
    if len(data) <= 1:
        return data
    mid = len(data) // 2
    left = merge_sort(data[:mid])
    right = merge_sort(data[mid:])
    return merge(left, right)

def merge(left, right):
    result = []
    i = j = 0
    while i < len(left) and j < len(right):
        if left[i] < right[j]:
                result.append(left[i])
                i += 1
        else:
                result.append(right[j])
                j += 1
```